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Class: BE Computer

Assignment No:03 - CNN Classifier — Fashion MNIST Dataset

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Framework: TensorFlow / Keras

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# Import libraries
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
import numpy as np

# -----
# Load Dataset (FASHION MNIST)
# -----
(x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load_data()

# -----
# Show Sample Images (like PDF)
# -----
plt.imshow(x_train[0])
plt.title("Sample Image 1")
plt.show()
plt.imshow(x_train[1])
plt.title("Sample Image 2")
plt.show()

# -----
# Print Shapes
# -----
print("x_train shape:", x_train.shape)
```

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print("x_test shape:", x_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)

# -----
# Data Preprocessing
# -----

x_train = x_train.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0
# Reshape to 4D for CNN
x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)

# -----
# Build CNN Model (same as PDF)
# -----

model = keras.Sequential([
    keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),
    keras.layers.Conv2D(64, (3,3), activation='relu'),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),
    keras.layers.Conv2D(128, (3,3), activation='relu'),
    keras.layers.Flatten(),
    keras.layers.Dense(128, activation='relu'),
    keras.layers.Dropout(0.25),
    keras.layers.Dense(10, activation='softmax')
])

# -----
# Compile Model

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# -----
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# -----
# Train Model
# -----

history = model.fit(x_train, y_train,
                   epochs=10,
                   validation_data=(x_test, y_test))

# -----
# Evaluate Model
# -----

test_loss, test_acc = model.evaluate(x_test, y_test)
print("Test Accuracy:", test_acc)

```

OUTPUT:



